Life Alongside AI

Course Syllabus

Welcome

Thank you for signing up to spend your spring break taking this course!

I think that one of the most important things we do as humans is contemplate difficult questions and that there are many ways one could do this. One could spend a lot of time studying a topic (otherwise known as a Bachelor's degree), one could create new knowledge in a topic (otherwise known as a PhD), one could explain the topic to others (becoming a communicator or lecturer), or one could engage in dynamic conversation with others, which is exactly the goal of this course. I believe that understanding our relationship with AI is going to be one of the most important questions we will grapple with in the coming decades, both individually and as a collective society, and I figured there was no better way to start doing this than in an intensive course format. There will be content I want you to learn, discussions I want you to participate in, and projects I want you to complete, but my real goal for this course is to have us deeply engage in our central question: how ought we to live alongside AI?

Course Information

Where and When

Classroom: HCRS 1066
Dates: Monday, March 16, 2026 – Thursday, March 20, 2026
Times: 8am-5pm

Instructor Information

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Prerequisites

None

Course Description

The increasing capabilities of artificial intelligence systems have caused humans to question the uniqueness of our contributions to knowledge work, creativity, and even relationships. To build a better understanding of our place in this emerging dyad, this course seeks to conduct a multiscale exploration of both human and artificial intelligence. We'll begin with the biological side, learning how cells use signaling pathways to process information and how these cellular levels processes build up to the functions of the human brain. In parallel, we'll explore how deep neural networks, which power today's most capable AI models, encode and transform information. We will use this multiscale lens to examine the relationship between biological and artificial information processing, and ask what it means to be a biologically intelligent being in a world of increasingly powerful artificial intelligence?

Course Objectives

Students will:

1. Explain the fundamentals of biological information processing using multiscale descriptions from molecular interactions to cellular signaling pathways to neural networks in the brain
2. Describe the basics of deep neural networks and the principles these networks use to encode information and make decisions
3. Identify unique and shared properties of information processing by comparing biological and artificial approaches
4. Apply a multiscale lens of information processing to evaluate the implications of artificial intelligence for human lives and livelihood
5. Engage in interdisciplinary dialogue bridging biology, computer science, and the humanities

Course Schedule

Day 1

The goal of Day 1 is to introduce the core concept of the course, a multiscale exploration of information processing in biological and digital systems, and then to dive into the world of biological information processing.

- Activity: Introductions
- Lecture: Course introduction and expectations
- Activity: Scaling and Emergence
- Lecture: Crash course in biological information processing
- Lecture: Crash course in Python coding
- Activity: Simulating a Neuron
- Project: Pick an example of biological intelligence and trace the behavior down to fundamental properties

Day 2

The goal of Day 2 is to explore the world of AI, starting with basic concepts and then eventually programming a simple neural network from the ground-up.

- Activity: Data reduction activity
- Lecture and Activity: understanding the basics of neural networks and building a network from scratch
- Reflection: Biological and Artificial Information Processing Comparison
- Lecture: connecting simple neural networks to LLMs
- Discussion: The “Algorithmic Doppelganger”

Day 3

The goal of Day 3 is to look at the history of human-machine relationships. How have we conceptualized and imagined machines in the past? What frameworks have we used to understand our place in the world?

- Activity: relationships to our own machines (phones, tablets, other)
- Lecture: A history of human-machine relational frameworks
- Debate: transhumanist vs critical posthumanism
- Project: Map different stories (movies, TV, books, etc) to different human-machine relational frameworks
- Reflection: Pull from any of the frameworks we have discussed and capture some aspect of your relationship to machines

Day 4

The goal of Day 4 is to generate answers to the central question of the course: how ought we, as beings with biological intelligence, to live in a world of increasingly powerful AI? The students will be given most of the day to explore that question in their desired way and format. The final assignment will be to answer that question, pulling on both the scaling/emergence themes discussed along with a framework for human-machine interaction.

- Activity: Respond to "Is AI Making Us Lazy?"
- Lecture: Project Expectations
- Final Project: Brainstorm
- Presentation: Project Pitches
- Final Project: Work Block
- Final Project: Presentations
- Reflection and Course Eval

Required Materials

- A laptop that can run a Python IDE

Assignments and Assessments

The graded assignments in the course breakdown are as follows:

Day	Category	Task	Points
Day 1	Reflection	Intro Activity	5
Day 1	Project	Biological Intelligence	10
Day 1	Participation	Introductions, Scaling and Emergence	5
Day 2	Project	Building a neural network	10
Day 2	Reflection	Comparing biological and artificial information processing	10
Day 2	Participation	The Algorithmic Doppelganger	5
Day 3	Participation	Debate: Transhumanist vs Critical Posthumanism	5
Day 3	Project	Story mapping	10
Day 3	Reflection	Apply framework	10
Day 4	Participation	Is AI Making Us Lazy	5
Day 4	Project	Final Project	15
Day 4	Reflection	Final Reflection	10

Grading Policy

My general philosophy towards grading in this class is that I will grade on two things:

1. Participation
2. Creative Thinking

Participation: You and the other students in the class will get the most out of the course when everyone participates in the discussions, reflections, and projects. You will be graded on your participation in class activities.

Creative Thinking: This course is designed to instigate deep discussion about the fundamental mechanics that make up our own intelligence and the functioning of Artificial Intelligence systems. I am not interested in what an LLM has to say about these topics – I am interested in what you have to say. I am looking to see you apply frameworks we learn in this class to new ideas through the projects.

The class grades are broken down by the following categories:

Category	Percentage
Projects	45%
Reflection Activities	35%
Class Participation	20%

Course Policies and Expectations

My biggest expectation is that you come to this course with curiosity and energy. We are here to probe nothing less than the fundamental questions of what it means to be human and how we ought to exist in a world with increasingly powerful AI. I have designed the course to cover a wide range of disciplines and activity types. I know not everything will resonate with everyone, but my hope is you walk in with an open mind to learn as much as you can in the time we have together.

This course is designed to be accessible to students from majors and disciplines, so I have no expectations around you knowing how to code. The knowledge and skills we will touch on along the way - biology, computer science, math, philosophy – are intended to be in service of interrogating the dyad between biological and artificial intelligence. I expect that you will become familiar with these topics, not experts.

Accessibility and Support Resources

This course is committed to providing an inclusive and accessible learning environment for all students. I recognize that students have diverse learning needs and am dedicated to ensuring everyone can fully participate and succeed in this course.

If you require accommodation due to a disability, chronic health condition, mental health need, or any other circumstance that may affect your learning, please contact me as soon as possible to discuss how I can best support you. All conversations will be kept confidential, and we will work together to implement appropriate accommodation.

Additionally, if you experience any challenges during the course—whether related to course materials, personal circumstances, technology access, or other concerns—please don't hesitate to communicate with me. The compressed nature of this spring break course means early communication is especially important so we can address any issues promptly.

I am here to help you succeed, and I encourage you to contact me to discuss any needs or concerns you may have.